

UTKARSH RAI

Boston, MA | rai.ut@northeastern.edu | +1-857-351-6420 | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Robotics MS with a focus on trajectory generation and planning using classical and deep learning-based methods for single and multi-agents. Peer-reviewed work in driving-scene perception (IROS'24) and multi-agent reinforcement learning (IEEE RA-L).

EDUCATION

Northeastern University

Master of Science in Robotics

Boston, MA

Expected Nov 2026

Coursework: Mobile Robotics, Robot Mechanics and Control, Computer Vision, Optimization

National Institute of Technology, Silchar

Bachelor of Technology in Electrical Engineering

Silchar, India

Coursework: Control Systems, Data Structures, Machine Learning, Operating Systems

TECHNICAL SKILLS

Languages: Python (fluent), C++ (intermediate), MATLAB

ML Frameworks & Libraries: PyTorch, TensorFlow, ONNX, TensorRT, CUDA, OpenCV

Robotics & Simulation: ROS, Isaac Gym, MuJoCo, Gazebo, Linux, Git

Methods: Trajectory prediction, behavior & motion planning, imitation learning, offline & multi-agent RL, sequence modeling, trajectory & numerical optimization, sim-to-real, model optimization

PROFESSIONAL EXPERIENCE

ManiroAI — Robotics Intern

Behavior modeling and policy learning for manipulation

May 2025 – Dec 2025

Boston, MA

- Developed conditional diffusion policies for multimodal trajectory generation, conditioning on perceptual (RGB-D / 3D) and proprioception.
- Built multimodal input pipelines fusing 3D point-cloud and proprioceptive features via cross-attention transformers and 3D encoders for robust policy learning.
- Applied offline RL (IQL, CQL) and behavior-regularized methods (TD3+BC) to learn policies from logged demonstration data, improving robustness with limited data.
- Generated synthetic training data through procedural simulation and domain randomization (visual + dynamics) in Isaac Gym, improving model generalization and sim-to-real transfer; validated gains via side-by-side hardware A/B trials.
- Built a leader-follower teleoperation data pipeline for scalable trajectory collection, capturing 1000+ hours of demonstration trajectories across simulation and real-world settings.

Helping Hands Lab, Northeastern University — Research Assistant

Sequence modeling and 3D scene representations for motion policies

Sept 2024 – Present

Boston, MA

- Implemented history-conditioned sequence models (long-context transformers / recurrent memory) for temporal prediction under partial observability, with latency-aware inference (chunking + caching).
- Built efficient 3D scene representations (Gaussian Splatting → point clouds) to support fast online motion evaluation and planning in dynamic environments.
- Optimized large-model rollouts with memory-efficient attention (FlashAttention, xFormers), reducing memory usage 2–3× to enable real-time policy execution.

Mowito — Robotics Engineer

Production trajectory generation and planning for warehouse automation

Sept 2022 – July 2024

Bangalore, India

- Deployed 50 Hz MPC-based trajectory generation and motion planners in C++ on UR5 hardware, improving pick-place throughput by 20%.
- Built a vision-to-control pipeline with stereo-depth pose estimation and MPC controllers for real-time picking.
- Developed pose estimation from RGB images and 3D voxel segmentation for optimal pick selection.

PUBLICATIONS

- U. Rai, S. Gangisetty, A. H. Abdul Hafez, A. Subramaniam, C. V. Jawahar. "Visual Place Recognition in Unstructured Driving Environments," IROS 2024. — **driving-scene perception**.
- W. Dai, U. Rai, J. Chiun, Y. Cao, G. Sartoretti. "Heterogeneous Multi-robot Task Allocation and Scheduling via Reinforcement Learning," IEEE RA-L. — **multi-agent coordination and interaction via RL**.